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Ch 3: Deep learning: Tensorflow and Keras

V1: Tensorflow and Keras Overview

Tensor $\rightarrow$ Vector / Data vector
Flow $\rightarrow$ Backpropagation.

Tensorflow is low-level lib.

Keras is high-level lib.

backend - could be - Tensorflow, Theano, Caffe, PyTorch, MXNET.

V2: GPU vs. CPU for Deep Learning

GPU - Graphics processing unit.

Sum of all cache in GPU is VideoRAM (VRAM).

V3: Google Laboratory

V4: code

Cross entropy = Average log loss

V7: MLP: Initialization

Before videos also in ipynb

V8: Model 1: Sigmoid Activation

V9: Model 2: ~~ReLU~~ ReLU activation

V10: Batch Normalization: Model 3

$$W^T x = z$$

Batch Normalize z and not x .

V11: Dropout: Model 4

V12: MNIST Classification in Keras

V13: Hyperparameter tuning in Keras

- 1) How many layers
- 2) # activation units in each layer
- 3) ReLU, Sigmoid
- 4) Dropout rate

Keras and Sklearn for hyperparameter-tuning.